

9 Problem-2

1) Solution

Pair of pure two-qubit state $|\psi\rangle$ and $|\phi\rangle$

Example-1

$$|\psi\rangle_{AB} = \frac{3}{\sqrt{10}}|00\rangle + \frac{1}{\sqrt{10}}|11\rangle$$

$$|\phi\rangle_{AB} = \frac{1}{\sqrt{10}}|00\rangle + \frac{3}{\sqrt{10}}|11\rangle$$

$$\lambda_\psi = (0.9, 0.1) \quad \left. \begin{array}{l} \\ \lambda_\phi = (0.7, 0.3) \end{array} \right\}$$

① $\lambda_\phi = 0.7 < \lambda_\psi = 0.9 \Rightarrow |\psi\rangle \xrightarrow{\text{local}} |\phi\rangle$

② $\lambda_\phi \neq \lambda_\psi \Rightarrow |\phi\rangle \not\xrightarrow{\text{local}} |\psi\rangle$

Example-2 - Pair of pure two-qubit state

$$|\psi\rangle_{AB} = \frac{2}{\sqrt{5}}|00\rangle + \frac{1}{\sqrt{5}}|11\rangle$$

$$|\phi\rangle_{AB} = \frac{2}{\sqrt{5}}|01\rangle + \frac{1}{\sqrt{5}}|10\rangle$$

$$= \frac{2}{\sqrt{5}}|10\rangle + \frac{1}{\sqrt{5}}|01\rangle$$

$\lambda_\psi > \lambda_\phi$ and $\lambda_\psi < \lambda_\phi \Rightarrow \lambda_\psi = \lambda_\phi$

where,

$$|\phi\rangle = (F \otimes X) |\psi\rangle$$

$$\left. \begin{array}{l} |\psi\rangle \xrightarrow{\text{local}} |\phi\rangle \\ |\phi\rangle \xrightarrow{\text{local}} |\psi\rangle \end{array} \right\} \lambda_\psi = \lambda_\phi$$

2)

Example $\rightarrow$ methane

$$|\psi\rangle_{AB} = \frac{1}{\sqrt{2}}|11\rangle + \sqrt{\frac{2}{5}}|22\rangle + \sqrt{\frac{1}{10}}|33\rangle$$

$$|\phi\rangle_{AB} = \sqrt{\frac{3}{5}}|11\rangle + \sqrt{\frac{1}{5}}|22\rangle + \frac{1}{\sqrt{5}}|33\rangle$$

$$A\phi = (0.5, 0.4, 0.1)$$

$$A\psi = (0.6, 0.2, 0.2)$$

So for first term, $(A\psi_1, A\phi_1)$

$$A\psi_1 = 0.5$$

$$A\phi_1 = 0.6$$

$$\left. \begin{array}{l} A\psi_1 = 0.5 \\ A\phi_1 = 0.6 \end{array} \right\} \Rightarrow A\psi_1 < A\phi_1$$

Apply the second term

$$0.5 + 0.4 = 0.9$$

$$0.6 + 0.2 = 0.8$$

$$\left. \begin{array}{l} 0.5 + 0.4 = 0.9 \\ 0.6 + 0.2 = 0.8 \end{array} \right\} \Rightarrow A\psi < A\phi$$

$$A\psi > A\phi \rightarrow \text{Satisfies}$$

So $A\psi > A\phi$

The first term

$$0.5 \neq 0.6 \text{ so } A\psi \neq A\phi$$

Note that:

Neither state can be converted other by LOCC

Nielsen's theorem

$$\left. \begin{array}{l} |\psi\rangle \xrightarrow{\text{LOCC}} |\phi\rangle \\ |\phi\rangle \xrightarrow{\text{LOCC}} |\psi\rangle \end{array} \right\}$$

2) Solution

No, Any pure two state is impossible neither direction at least work in one direction.

$$|\psi\rangle = \sqrt{a} |00\rangle + \sqrt{1-a} |11\rangle$$

Schmidt Vector

$$\left. \begin{aligned} \lambda_\psi &= (a, 1-a) \\ \lambda_\phi &= (a, 1-a) \end{aligned} \right\} \text{ where } a, a \in [\frac{1}{2}, 1]$$

Weyl's Theorem Says

$$|\psi\rangle \xrightarrow{\text{hocc}} |\phi\rangle \Leftrightarrow \lambda_\psi \succ \lambda_\phi$$

Majorization

$$(a, 1-a) \succeq (a, 1-a) \Leftrightarrow a \geq a$$

Now for any two numbers a and a is either

$$a \geq a \text{ so } |\psi\rangle \xrightarrow{\text{hocc}} |\phi\rangle$$

$$a \geq a \text{ so } |\phi\rangle \xrightarrow{\text{hocc}} |\psi\rangle$$

Impossible neither direction of two possible

3) Solution

The PVM measurement, where U is unitary operator,

$$\text{where, } U = \frac{1}{\sqrt{N}} \sum_{i=1}^N \sqrt{p_i} |i\rangle \langle i|, \quad I = \sum_{i=1}^N |i\rangle \langle i| \quad 3)$$

$$|\phi^+\rangle_{AB} \xrightarrow{\text{h.o.c.c.}} |\psi\rangle_{AB}$$

Therefore

$$|\psi\rangle_{AB} = (U \otimes I) |\phi^+\rangle_{AB}$$

Apply operators on the Alice subit

$$|\psi\rangle_{AB} = (U \otimes I) |\phi^+\rangle_{AB}$$

$$= (U \otimes I) \frac{1}{\sqrt{n}} \sum_{i=1}^n |i\rangle_A |i\rangle_B$$

$$= \frac{1}{\sqrt{n}} \sum_{i=1}^n (U|i\rangle_A \otimes I|i\rangle_B)$$

$$= \frac{1}{\sqrt{n}} \sum_{i=1}^n \left[\sqrt{p(i)} \sum_{j=1}^n |i\rangle_A \langle j| \otimes \sum_{k=1}^n |j\rangle_B \langle k| \right]$$

$$= \frac{\sqrt{n}}{\sqrt{n}} \sum_{i=1}^n \sqrt{p(i)} \left[\sum_{j=1}^n |i\rangle_A \langle j| \otimes \sum_{k=1}^n |j\rangle_B \langle k| \right]$$

$$= \sum_{i=1}^n \sqrt{p(i)} [|i\rangle_A \otimes |i\rangle_B]$$

$$= \sum_{i=1}^n \sqrt{p(i)} |i\rangle_{AB}$$

$$= |\psi\rangle_{AB}$$

$$\underline{\underline{|\psi\rangle_{AB} = |\psi\rangle_{AB}}}$$

A)

4) Solution

$$|\psi\rangle = \sqrt{0.4}|00\rangle + \sqrt{0.4}|11\rangle + \sqrt{0.2}|12\rangle + \sqrt{0.2}|13\rangle$$

$$|\psi\rangle = \sqrt{0.5}|00\rangle + \sqrt{0.25}|11\rangle + \sqrt{0.25}|12\rangle$$

Schmidt Vector

$$\lambda\psi = (0.4, 0.4, 0.2, 0.2)$$

$$\lambda\phi = (0.5, 0.25, 0.25, 0)$$

Therefore

$$k=2 \rightarrow \begin{cases} \lambda\psi_2 = 0.4 \\ \lambda\phi_2 = 0.5 \end{cases} \Rightarrow \lambda\psi_2 < \lambda\phi_2$$

$$k=2 \rightarrow \begin{cases} \lambda\psi_2 = 0.4 + 0.4 = 0.8 \\ \lambda\phi_2 = 0.5 + 0.25 = 0.75 \end{cases} \left. \begin{array}{l} \lambda\psi_2 \neq \lambda\phi_2 \text{ so} \\ \lambda\psi_2 > \lambda\phi_2 \end{array} \right\}$$

fails

The maximization condition

$$\lambda\psi \rightarrow \lambda\phi \text{ fails at } k=2$$

$$|\psi\rangle \xrightarrow{\text{HCC}} |\phi\rangle$$

where, Alice and Bob

Show Carol's work

$$|\psi\rangle|c\rangle \longrightarrow |\phi\rangle|c\rangle$$

Entangled pair of qubit in the state

$$|c\rangle = \sqrt{0.6}|00\rangle + \sqrt{0.4}|11\rangle \Rightarrow \lambda(c) = (0.6, 0.4)$$

Product State, Schmidt Vectors tensor multiply

$$\lambda(\psi \otimes \phi) = \lambda(\psi) \otimes \lambda(\phi)$$

$$\lambda(\phi \otimes \psi) = \lambda(\phi) \otimes \lambda(\psi)$$

compute $\lambda(\psi \otimes \phi)$

All products:

$$\{0.4, 0.4, 0.1, 0.1\} \times \{0.6, 0.4\}$$

$$= (0.24, 0.16, 0.24, 0.16, 0.06, 0.04, 0.06, 0.04)$$

$$\lambda(\psi \otimes \phi) = (0.24, 0.16, 0.24, 0.16, 0.06, 0.04, 0.06, 0.04)$$

compute

$$\lambda(\phi \otimes \psi) = (0.5, 0.25, 0.25, 0) \times (0.6, 0.4)$$

$$= (0.30, 0.20, 0.15, 0.20, 0.15, 0.20, 0, 0)$$

majorization

$$\lambda(\psi \otimes \phi) \succ \lambda(\phi \otimes \psi)$$

Compare each k - every partial sum of $\psi \otimes \phi$ is $\leq$ the corresponding one of $\phi \otimes \psi$

Cumulative sum

$$\text{for } \psi \otimes \phi = (0.24, 0.48, 0.64, 0.80, 0.86, 0.92, 0.96, 1)$$

$$\text{for } \phi \otimes \psi = (0.3, 0.5, 0.65, 0.8, 0.9, 1, 1, 1)$$

6)

Therefore

$$\left. \begin{aligned} 0.48 &\leq 0.50 \\ 0.64 &\leq 0.65 \\ 0.92 &\leq 1.00 \end{aligned} \right\}$$

Therefore,

$$\chi(\psi \otimes C) \rightarrow \chi(\phi \otimes C)$$

NP's Theorem

$$|\psi\rangle|C\rangle \xrightarrow{\text{HCC}} |\phi\rangle|C\rangle$$

Problem-2, Entanglement monogamy

2) Solution

$$\begin{aligned} |\psi\rangle_{ABC} &= |00\rangle_{AB} \otimes (\alpha_0|0\rangle_C + \alpha_1|1\rangle_C) + |11\rangle_{AB} \otimes (\beta_0|0\rangle_C + \beta_1|1\rangle_C) \\ &= |00\rangle_{AB} \otimes (\alpha_0|0\rangle_C + \alpha_1|1\rangle_C) + |11\rangle_{AB} \otimes (\beta_0|0\rangle_C + \beta_1|1\rangle_C) \end{aligned}$$

Q1) Solution

$$|0\rangle = \frac{1}{\sqrt{2}}(|+\rangle + |-\rangle)$$

$$|1\rangle = \frac{1}{\sqrt{2}}(|+\rangle - |-\rangle)$$

$$|00\rangle_{AB} = \left[\frac{1}{\sqrt{2}}(|+\rangle + |-\rangle) \otimes \frac{1}{\sqrt{2}}(|+\rangle + |-\rangle) \right]_{AB}$$

$$= \frac{1}{2} [|++\rangle_{AB} + |+-\rangle_{AB} + |-+\rangle_{AB} + |--\rangle_{AB}]$$

$$|11\rangle_{AB} = \frac{1}{\sqrt{2}} [|+\rangle - |-\rangle]_A \otimes \frac{1}{\sqrt{2}} [|+\rangle - |-\rangle]_B$$

$$= \frac{1}{2} [1+\gamma - 1+\gamma - 1+\gamma + 1-\gamma]_{AB}$$

Express in the X-basis

$$\begin{aligned} |\psi\rangle_{ABC} &= \frac{1}{2} [1+\gamma + 1+\gamma + 1+\gamma + 1-\gamma]_{AB} \otimes (\alpha_0 |0\rangle_C + \alpha_1 |1\rangle_C) \\ &\quad + \frac{1}{2} [1+\gamma - 1+\gamma - 1+\gamma + 1-\gamma]_{AB} \otimes (\beta_0 |0\rangle_C + \beta_1 |1\rangle_C) \end{aligned}$$

$$\begin{aligned} &= \frac{1}{2} [1+\gamma]_{AB} ((\alpha_0 + \beta_0) |0\rangle_C + (\alpha_1 + \beta_1) |1\rangle_C) \\ &\quad + 1+\gamma]_{AB} ((\alpha_0 - \beta_0) |0\rangle_C + (\alpha_1 - \beta_1) |1\rangle_C) + \\ &\quad 1-\gamma]_{AB} ((\alpha_0 - \beta_0) |0\rangle_C + (\alpha_1 - \beta_1) |1\rangle_C) + \\ &\quad 1-\gamma]_{AB} ((\alpha_0 + \beta_0) |0\rangle_C + (\alpha_1 + \beta_1) |1\rangle_C) \end{aligned}$$

b) Solve for

$$\begin{aligned} |\psi\rangle_C &= \alpha_0 |0\rangle + \alpha_1 |1\rangle \\ |\psi\rangle_C &= \beta_0 |0\rangle + \beta_1 |1\rangle \end{aligned}$$

and,

$$\begin{aligned} |\psi\rangle_{ABC} &= \frac{1}{2} [1+\gamma]_{AB} [\alpha_0 |0\rangle_C + \beta_0 |0\rangle_C + \alpha_1 |1\rangle_C + \beta_1 |1\rangle_C] + \\ &\quad 1+\gamma]_{AB} [\alpha_0 |0\rangle_C - \beta_0 |0\rangle_C + \alpha_1 |1\rangle_C - \beta_1 |1\rangle_C] + \\ &\quad 1-\gamma]_{AB} [\alpha_0 |0\rangle_C - \beta_0 |0\rangle_C + \alpha_1 |1\rangle_C - \beta_1 |1\rangle_C] + \\ &\quad 1-\gamma]_{AB} [\alpha_0 |0\rangle_C + \beta_0 |0\rangle_C + \alpha_1 |1\rangle_C + \beta_1 |1\rangle_C] \end{aligned}$$

$$= \frac{1}{2} [1 + \gamma_{AB} \underbrace{[\alpha_0 | 0 \rangle_c + \alpha_1 | 1 \rangle_c]}_{= | \psi_\alpha \rangle_c} + \underbrace{[\beta_0 | 0 \rangle_c + \beta_1 | 1 \rangle_c]}_{= | \psi_\beta \rangle_c}] +$$

$$1 + \gamma_{AB} \underbrace{[\alpha_0 | 0 \rangle_c + \alpha_1 | 1 \rangle_c]}_{= | \psi_\alpha \rangle_c} - \underbrace{[\beta_0 | 0 \rangle_c + \beta_1 | 1 \rangle_c]}_{= | \psi_\beta \rangle_c}] +$$

$$1 - \gamma_{AB} \underbrace{[\alpha_0 | 0 \rangle_c + \alpha_1 | 1 \rangle_c]}_{= | \psi_\alpha \rangle_c} - \underbrace{[\beta_0 | 0 \rangle_c + \beta_1 | 1 \rangle_c]}_{= | \psi_\beta \rangle_c}] +$$

$$1 - \gamma_{AB} \underbrace{[\alpha_0 | 0 \rangle_c + \alpha_1 | 1 \rangle_c]}_{= | \psi_\alpha \rangle_c} + \underbrace{[\beta_0 | 0 \rangle_c + \beta_1 | 1 \rangle_c]}_{= | \psi_\beta \rangle_c}]$$

$$\begin{aligned} | \psi \rangle_{ABC} &= \frac{1}{2} [1 + \gamma_{AB} (| \psi_\alpha \rangle_c + | \psi_\beta \rangle_c) + 1 + \gamma_{AB} (| \psi_\alpha \rangle_c - | \psi_\beta \rangle_c) \\ &\quad + 1 - \gamma_{AB} (| \psi_\alpha \rangle_c - | \psi_\beta \rangle_c) + 1 - \gamma_{AB} (| \psi_\alpha \rangle_c + | \psi_\beta \rangle_c)] \\ &= \frac{1}{2} [1 + \gamma_{AB} (| \psi_\alpha \rangle_c + | \psi_\beta \rangle_c) + 1 + \gamma_{AB} (| \psi_\alpha \rangle_c - | \psi_\beta \rangle_c) \\ &\quad + 1 - \gamma_{AB} (| \psi_\alpha \rangle_c - | \psi_\beta \rangle_c) + 1 - \gamma_{AB} (| \psi_\alpha \rangle_c + | \psi_\beta \rangle_c)] \end{aligned}$$

Probabilities to be zero we must have

$$| \psi_\alpha \rangle - | \psi_\beta \rangle = 0 \quad \dots \quad (x)$$

- The expansion above the amplitudes of $1 + \gamma$ and $1 - \gamma$ are proportional to

$$| \psi_\alpha \rangle - | \psi_\beta \rangle = 0 \Rightarrow | \psi_\alpha \rangle = | \psi_\beta \rangle = | \psi \rangle_c \quad (y)$$

Ex 8.9.10

$$\begin{cases} a_0 = p_0 \\ a_1 = p_1 \end{cases}$$

$$147ABC = \frac{1}{2} [1+7AB (147C + 147C) + 1+7AB [0] + 1+7AB [0] \\ + 1-7AB [147C + 147C]]$$

$$= \frac{1}{2} [1+7AB [2147C] + 1-7AB [2147C]]$$

$$= \frac{1}{2} [1+7AB + 1-7AB] 2147C$$

$$= \frac{2}{2} [1+7AB + 1-7AB] \otimes 147C$$

$$= [1+7AB + 1-7AB] \otimes 147C$$

$$= (1007 + 1227)AB$$

$$= \frac{\sqrt{2}(1007 + 1227)AB \otimes 147C}{\sqrt{2}}$$

$$= \sqrt{2} \left[\frac{1007 + 1227}{\sqrt{2}} \right] AB \otimes 147C \\ = 1477$$

$$= \sqrt{2} 1477AB \otimes 147C //$$

201

$$|\psi\rangle_{ABC} = \frac{1}{\sqrt{2}} |\phi\rangle_{AB} \otimes |\psi\rangle_C$$

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2) Solution

No the global pure state $|\psi\rangle_{ABC}$ cannot be entangled b/w AB and C. It must be a product across the cut AB|C.

Given

$|\phi\rangle_{AB} \rightarrow$ entangled pure state }
 $|\psi\rangle_C \rightarrow$ another state }

$$|\psi\rangle_{ABC} = |\psi\rangle_{AB} \otimes |\psi\rangle_C$$

$$\rho_{ABC} = |\psi\rangle_{ABC} \langle \psi|_{ABC}$$

$$= |\psi\rangle_{AB} \otimes |\psi\rangle_C \langle \phi|_{AB} \otimes \langle \psi|_C$$

where,

$$\rho_{AB} = \text{Tr}_C [\rho_{ABC}]$$

$$= \text{Tr}_C [|\psi\rangle_{AB} \otimes |\psi\rangle_C \langle \phi|_{AB} \otimes \langle \psi|_C]$$

$$= |\psi\rangle_{AB} \langle \phi|_{AB} \underbrace{\text{Tr}_C [|\psi\rangle_C \langle \psi|_C]}_{=1}$$

$$= |\psi\rangle_{AB} \langle \phi|_{AB}$$

h21

$$= \frac{1}{3} [\underbrace{|01\rangle|10\rangle + |10\rangle|10\rangle + |10\rangle|01\rangle + |10\rangle|10\rangle + |00\rangle|00\rangle}_{= 2|\psi^+\rangle|1\rangle}]$$

$$= \frac{2}{3} |\psi^+\rangle|1\rangle + \frac{1}{3} |00\rangle|00\rangle$$

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$$S_{AB}^{GHZ} = \text{Tr}_C [|GHZ\rangle\langle GHZ|]$$

$$= \text{Tr}_C [\frac{1}{2} [|000\rangle\langle 000| + |000\rangle\langle 111| + |111\rangle\langle 000| + |111\rangle\langle 111|]]$$

$$= \frac{1}{2} [|00\rangle_{AB}\langle 00|_{AB} \text{Tr}_C [|0\rangle\langle 0| + 0 + 0 + |1\rangle\langle 1|] + \text{Tr}_C [|1\rangle\langle 1|]]$$

$$= \frac{1}{2} |00\rangle_{AB}\langle 00|_{AB} + \frac{1}{2} |11\rangle_{AB}\langle 11|_{AB}$$

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To compare, "How much Entanglement?"

1) Concurrence,

$$\left. \begin{aligned} C(S_{AB}^{10}) &= \frac{2}{3} \\ C(S_{AB}^{GHZ}) &= 0 \end{aligned} \right\}$$

13)
12)

Concursion / monogamy Interpretation

① more distributed pairwise entanglement: W state

W
 S_{AB} is entangled,

② more collective entanglement GHZ state.

Two qubit reduced state

GHZ
 S_{AB} is separable

so, W spreads entanglement among pairs

GHZ - concentrates it as all-or-nothing three-way